

Criticality of Demorgan formulas

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1 Introduction

- Demorgan formulas
- Criticality
- Applications of Criticality

Theorem 1. *Let F be a DeMorgan formula whose number of leaves is $L(F)$. Then there exists an absolute constant c such that*

$$\Pr_{\rho \sim \mathcal{R}_p} [\text{Dt}_{\text{depth}}(F|_{\rho}) \geq t] \leq (p\sqrt{L(F)})(c \cdot p\sqrt{L(F)})^{t-1}$$

fix constants

2 Introduction

2.1 Decision trees

2.2 Random restrictions

2.3 Organization

3 Preliminaries

3.1 Canonical decision tree

In this section, we give an algorithm CDT for constructing a canonical decision tree for a DeMorgan formula F w.r.t a restriction ρ .

We begin with a subroutine called *Next variable function* NVF which tells us which variable to query next given a formula a restriction

NVF(Next Variable Function)

Input: Demorgan formulas F and a restriction ρ .

Output: a leaf variable of F .

1. If $F|_{\rho} \equiv 0/1$, return $\perp$.
2. If F is a variable say x_i such that $\rho(x_i) = *$, return x_i .
3. If $F = F_1 \vee F_2$:
 - If $F_1|_{\rho} \equiv 0$:
return NVF (F_2, ρ)
 - else:
return NVF (F_1, ρ)

try putting
it in an algo-
rithm env

CDT: Input: DeMorgan formula F and a restriction ρ . Output: A decision tree for $F|_{\rho}$.

1. If $F|_{\rho} \equiv 0/1$, return 0/1 accordingly.

2 Otherwise: call NVF (F, ρ)
 If NVF $(F, \rho) = x_j$, query x_j and
 If x_j is 0,
 return CDT $(F, \rho \cup \{x_j \mapsto 0\})$
 If x_j is 1,
 return CDT $(F, \rho \cup \{x_j \mapsto 1\})$.

3.2 Communication game

Given a DeMorgan formula F and collection of restrictions $\mathcal{R}$ such that $F|_\rho \neq \text{constant } \forall \rho \in \mathcal{R}$ and a bit $a \in \{0, 1\}$, we can define game $\text{Game}_{F, \mathcal{R}, a}$ as follows:

Alice and Bob get the partial inputs $\rho(i \mapsto a)$ and $\rho(i \mapsto \bar{a})$ respectively where, $\rho \in \mathcal{R}$, $a \in \{0, 1\}$ and $\text{NVF}(F, \rho) = x_i$. Alice and Bob know $F, \mathcal{R}$ and a but they don't know ρ . Their goal is to output the variable where their partial inputs differ.

3.2.1 Protocols and Rectangles

Similar to the Karchmer Wigderson games, a communication protocol can be described using a tree whose internal nodes are labelled Alice or Bob and the leaves are labelled with the input variables of F . A *transcript* is a string of messages exchanged between Alice and Bob on a given set of inputs.

We use Π to denote the *protocol*, $\Pi(x, y)$ to denote the output of the protocol on inputs x, y and Π_{xy} to denote the transcript of the protocol Π on inputs x, y .

For a the game $\text{Game}_{F, \mathcal{R}, a}$, we study its *communication matrix* denoted by $M_{F, \mathcal{R}, a}$ ¹. Whose rows are indexed by the set of Alice's inputs X and columns are indexed by the set of Bob's inputs Y . For any $x \in X$ and $y \in Y$, $M(x, y) = \Pi(x, y)$.

$X = \{\rho \circ (\text{NVF}(F, \rho) \rightarrow a) \mid \rho \in \mathcal{R}\}$ and $Y = \{\rho \circ (\text{NVF}(F, \rho) \rightarrow \bar{a}) \mid \rho \in \mathcal{R}\}$.

A *Rectangle* is an object of the form $A \times B$ for some $A \in X$ and $B \in Y$.

Similar to KW games, a protocol induces a partition of the communication matrix into monochromatic rectangles. Where the rectangles in the partition correspond to leaves of the protocol.

Lemma 2. *Let p, q be distributions supported on X, Y respectively. For any transcript π , let $A_\pi \times B_\pi$ be the corresponding mono-chromatic rectangle. Then,*

$$\sum_{\pi \in \Pi} p(A_\pi) \cdot q(B_\pi) = 1$$

write well

4 Protocol

Lemma 3. *For every DeMorgan formula F , there exist a protocol that achieves the above guarantee with at most $O(L(F)^2)$ leaves.*

Proof. Alice gets $\rho_j = \rho \circ \sigma_j$ and Bob gets $\rho'_j = \rho \circ \sigma'_j$ and they want figure out the variable where their inputs differ.

$CC_F(\rho_j, \rho'_j)$ Suppose $F = F_1 \vee F_2$,

Alice sends 0 if $F_1|_{\rho_j} \equiv 0$ and 1 otherwise.

If Alice sends 1,

* then both of them run $CC_{F_1}(\rho_j, \rho'_j)$

If Alice sends 0,

* Bob checks if $F_1|_{\rho'_j} \equiv 0$ and sends a 0 if that is the case and 1 otherwise.

* If Bob sends a 0,

* they run $CC_{F_2}(\rho_j, \rho'_j)$ and

* otherwise

¹we some times drop the subscripts from $M_{F, \mathcal{R}, a}$ for aesthetic reasons if $F, \mathcal{R}, a$ are clear from the context.

* Bob arbitrarily extends ρ'_j to a 1 input ρ' they run $CC_{F_1}(\rho_j, \rho')$.

*Need to argue the correctness of the protocol.

(We can assume that at each node of the formula tree, the left child is smaller than the right child.)

Claim 4 (Complexity). *The number of transcripts in the above protocol is atmost $L(F)^2$.*

Let $T(F)$ denote the number of transcripts of the protocol on for F . We have the recursion,

$$\begin{aligned} T(F) &\leq T(F_1) + T(F_2) + L(F_1) \\ &\leq T(F_1) + T(F_2) + L(F)/2 \\ \implies T(F) &= O(L(F)^2) \end{aligned}$$

□

4.1 Notation

Let $\phi_a(F, \rho) = \rho(\text{NVF}(F, \rho) \mapsto a)$. Since F is usually clear from the context, we just write $\phi_a(\rho)$.
 $\phi_{a_1, \dots, a_t}^t(\rho) = \phi_{a_t}(\phi_{a_1, \dots, a_{t-1}}^{t-1}(\rho))$.

5 Proof

Lemma 5.

$$\Pr_{\rho} [\text{CDT}(F, \rho) \text{ has depth } t] \leq (p\sqrt{L(F)})(c \cdot p\sqrt{L(F)})^{t-1}$$

Proof. If $\text{CDT}(F, \rho)$ has depth t , then there exist $a \in \{0, 1\}^t$ and variables $x_{i_1}, \dots, x_{i_t}$ such that for all $k \in [t]$, $\text{NVF}(F, \rho \circ \sigma_{k-1}) = x_{i_k}$, where, $\sigma_k = (x_{i_1} \rightarrow a_1, \dots, x_{i_k} \rightarrow a_k)$, $\sigma'_k = (x_{i_1} \rightarrow a_1, \dots, x_{i_k} \rightarrow \bar{a}_k)$ $\forall k \in [t]$ and $\sigma_0 = \sigma'_0 = ()$.

Let $X^{(t)} = \{\phi_{a_t}(\rho) : F|_{\phi_{a_t}(\rho)} \equiv 0, F|_{\phi_{\bar{a}_t}(\rho)} \equiv 1\}$ $Y^{(t)} = \{\phi_{\bar{a}_t}(\rho) : F|_{\phi_{a_t}(\rho)} \equiv 0, F|_{\phi_{\bar{a}_t}(\rho)} \equiv 1\}$.

$X^{(k)} = \{\phi_{a_k}(\rho) : \Pi^{(k)}(\phi_{a_k}(\rho), \phi_{\bar{a}_k}(\rho)) = \ell_k, \Pi^{(j)}(\phi_{a_k}(\rho) \circ \sigma_{k+1 \rightarrow j}, \phi_{a_j}(\rho) \circ \sigma'_{k+1 \rightarrow j}) = \ell_j, F|_{\phi_{a_k}(\rho) \circ \sigma_{k+1 \rightarrow t}} \equiv 0, F|_{\phi_{a_k}(\rho) \circ \sigma'_{k+1 \rightarrow t}} \equiv 1\}$

Let $\mathcal{R}_k = \{\rho \circ \sigma_{k-1} : \Pi^{(j)}(\rho \circ \sigma_j, \rho \circ \sigma'_j) = \ell_j \forall 1 \leq j \leq k-1\}$ and $\mathcal{R}_1 = \{\rho : \text{NVF}(F, \rho) \neq \perp\}$. Consider the games $\mathcal{G}_k$ for each $k \in [t]$ where $\mathcal{G}_k = \text{Game}_{F, \mathcal{R}_k, a_k}$. [Tulasi: is $\mathcal{R}_k$ this or from below?] By Lemma ??, there exists a protocol $\Pi^{(k)}$ for the game $\mathcal{G}_k$ for each $k \in [t]$ whose number of transcripts is $|\Pi^{(k)}|$. Note that $\mathcal{G}_k, \Pi^{(k)}$ depend only the variables $x_{i_1}, \dots, x_{i_{k-1}}$.

$$\begin{aligned} \Pr_{\rho} [\text{CDT}(F, \rho) \text{ has depth } t] &\leq \sum_{x_{i_1}, \dots, x_{i_t}} \Pr_{\rho} \left[\begin{array}{l} \forall 1 \leq j \leq t \\ (i) \text{NVF}(F, \rho \circ \sigma_{j-1}) = i_j \\ (ii) \rho(i_j) = * \\ (iii) F|_{\rho \circ \sigma_t} \equiv 0, F|_{\rho \circ \sigma'_t} \equiv 1 \end{array} \right] \\ &\quad \text{(replacing NVF with Protocol)} \\ &= \sum_{x_{i_1}, \dots, x_{i_t}} \sum_{\ell_1, \dots, \ell_t} \Pr_{\rho} \left[\begin{array}{l} \forall 1 \leq j \leq t \\ (i) \Pi^{(j)}(\rho \circ \sigma_j, \rho \circ \sigma'_j) = \ell_j \\ (ii) \rho(i_j) = * \\ (iii) F|_{\rho \circ \sigma_t} \equiv 0, F|_{\rho \circ \sigma'_t} \equiv 1 \end{array} \right] \\ &\quad \text{(substituting } \rho \circ \sigma_t \text{ with } \rho \text{ and rewriting the above event)} \\ &= \left(\frac{2p}{1-p} \right)^t \sum_{x_{i_1}, \dots, x_{i_t}} \sum_{\ell_1, \dots, \ell_t} \Pr_{\rho} \left[\begin{array}{l} \forall 1 \leq j \leq t \\ (i) \Pi^{(k)}(\rho(i_{j+1}, \dots, i_t = *), \rho(i_j \rightarrow \bar{a}_j, i_{j+1}, \dots, i_t = *)) = \ell_j \\ (ii) \rho(i_j) = a_j \\ (iii) F|_{\rho} \equiv 0, F|_{\rho(i_t \rightarrow a_t)} \equiv 1 \end{array} \right] \end{aligned}$$

The above equality follows from the following observations

- ρ can be recovered from $\rho \circ (i_1 \rightarrow a_1, \dots, i_t \rightarrow a_t)$ given $(i_1, \dots, i_t)$ and $a \in \{0, 1\}^t$.

- $\frac{\Pr[\rho]}{p^t} = \frac{\Pr[\rho \circ \sigma_t]}{((1-p)/2)^t}$ when $\rho(i_1) = \dots = \rho(i_t) = *$.

$$\begin{aligned}
& \Pr_{\rho} \left[\begin{array}{l} \forall 1 \leq j \leq t \\ (i) \Pi^{(k)}(\rho(i_{j+1}, \dots, i_t = *), \rho(i_j \rightarrow \bar{a}_j, i_{j+1}, \dots, i_t = *)) = \ell_j \\ (ii) \rho(i_j) = a_j \\ (iii) F|_{\rho \equiv 0, F|_{\rho(i_t \rightarrow a_t)} \equiv 1} \end{array} \right] \\
&= \prod_{k=t}^1 \Pr_{\rho} \left[\begin{array}{l} \forall k \leq j \leq t \\ (i) \Pi^{(k)}(\rho(i_{j+1}, \dots, i_t = *), \rho(i_j \rightarrow \bar{a}_j, i_{j+1}, \dots, i_t = *)) = \ell_j \\ (ii) \rho(i_j) = a_j \\ (iii) F|_{\rho \equiv 0, F|_{\rho(i_t \rightarrow a_t)} \equiv 1} \end{array} \right] \left[\begin{array}{l} \forall k+1 \leq j \leq t \\ (i) \Pi^{(k)}(\rho(i_{j+1}, \dots, i_t = *), \rho(i_j \rightarrow \bar{a}_j, i_{j+1}, \dots, i_t = *)) = \ell_j \\ (ii) \rho(i_j) = a_j \\ (iii) F|_{\rho \equiv 0, F|_{\rho(i_t \rightarrow a_t)} \equiv 1} \end{array} \right] 2 \\
&\quad \text{(factorizing the event into conditional events)} \\
&= \prod_{k=t}^1 \Pr_{\rho} \left[\begin{array}{l} \rho \in E_{k+1} \\ \rho(i_k) = a_k \\ \Pi^{(k)}(\rho(i_{k+1}, \dots, i_t = *), \rho(i_k \rightarrow \bar{a}_k, i_{k+1}, \dots, i_t = *)) = \ell_k \end{array} \right] \Bigg/ \Pr_{\rho} [\rho \in E_{k+1}] \\
&\quad \text{(E_k is shorthand for the event)} \\
&= \prod_{k=t}^1 \frac{\sum_{\rho \in F_{k+1}, \rho(i_k) = a_k} \Pr[\rho]}{\frac{\Pi^{(k)}(\rho, \rho(i_k \rightarrow \bar{a}_k)) = \ell_k}{\sum_{\rho \in F_{k+1}} \Pr[\rho]}} \\
&\quad \text{(substituting } E_k \text{ } \rho(i_{k+1}, \dots, i_t = *) \text{ with } \rho \text{ } F_k) \\
&\leq \prod_{k=t}^1 \sqrt{\frac{\sum_{\rho \in F_{k+1}, \rho(i_k) = a_k} \Pr[\rho]}{\frac{\Pi^{(k)}(\rho, \rho(i_k \rightarrow \bar{a}_k)) = \ell_k}{\sum_{\rho \in F_{k+1}} \Pr[\rho]}}} \sqrt{\frac{\sum_{\rho \in F_{k+1}, \rho(i_k) = *} \Pr[\rho]}{\frac{\Pi^{(k)}(\rho(i_k \rightarrow a_k), \rho) = \ell_k}{\sum_{\rho \in F_{k+1}} \Pr[\rho]}}} \\
&\quad \text{[Tulasi :this is a crucial step. Check.]} \\
&= \sqrt{\frac{\Pr[A_k(\ell_k)]}{\Pr[X_k]}} \sqrt{\frac{\Pr[B_k(\ell_k)]}{\Pr[Y_k]}}
\end{aligned}$$

Where, $E_k = \{\rho : \Pi^{(k)}(\rho(i_{j+1}, \dots, i_t = *), \rho(i_j \rightarrow \bar{a}_j, i_{j+1}, \dots, i_t = *)) = \ell_j, \rho(i_j) = a_j \forall k \leq j \leq t\}$
 $F_{k+1} = \{\rho : \Pi^{(j)}(\rho \circ \sigma_{k+1 \rightarrow j}, \rho \circ \sigma'_{k+1 \rightarrow j}), \rho(i_j) = * \forall k+1 \leq j \leq t\}$
 $X^{(t)} = \{\phi_{a_t}(\rho) : F|_{\phi_{a_t}(\rho)} \equiv 0, F|_{\phi_{a_t}(\rho)} \equiv 1\}$
 $Y^{(t)} = \{\phi_{\bar{a}_t}(\rho) : \rho \in X_{t+1}, F|_{\phi_{a_t}(\rho)} \equiv 0, F|_{\phi_{\bar{a}_t}(\rho)} \equiv 1\}$
 $X^{(k)} = \{\phi_{a_k}(\rho) : \rho \in \mathcal{A}_{k+1}\}$
 $Y^{(k)} = \{\phi_{\bar{a}_k}(\rho) : \rho \in \mathcal{A}_{k+1}\}$
 $A^{(k)}(\ell_k) = \{\rho : \rho \in X^{(k)}, \Pi^{(k)}(\rho, \rho(i_k \rightarrow \bar{a}_k)) = \ell_k\}$
 $B^{(k)}(\ell_k) = \{\rho : \rho \in Y^{(k)}, \Pi^{(k)}(\rho(i_k \rightarrow a_k), \rho) = \ell_k\}$

add skipped steps

$$\begin{aligned}
\sum_{x_{i_k}} \sum_{\ell_k} \sqrt{\frac{\Pr[A_k(\ell_k)]}{\Pr[X_k]}} \sqrt{\frac{\Pr[B_k(\ell_k)]}{\Pr[Y_k]}} &\leq \sqrt{|\Pi^{(k)}|} \sqrt{\sum_{x_k, \pi_k} \frac{\Pr[A_k(\ell_k)]}{\Pr[X_k]} \frac{\Pr[B_k(\ell_k)]}{\Pr[Y_k]}} \\
&= \sqrt{|\Pi^{(k)}|}
\end{aligned}$$

Therefore, $\Pr_{\rho} [\text{CDT}(F, \rho) \text{ has depth } t] \leq p^t \prod_{k=1}^t \sqrt{|\Pi^{(k)}|}$

□

$$\begin{aligned}
\Pr_{\rho} [\text{CDT}(F, \rho) \text{ has depth } 2] &\leq \sum_{x_{i_1}, x_{i_2}} \sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \Pr[\rho] \\
&= \sum_{x_1, x_2} \sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho] \cdot \frac{\sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \Pr[\rho]}{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho]}
\end{aligned}$$

$$\begin{aligned}
\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho] &= O(p) \sum_{\text{NVF}(F, \rho) = i_1} \sqrt{\Pr[\rho(i_1 \rightarrow a_1)] \Pr[\rho(i_1 \rightarrow \bar{a}_1)]} \\
&\leq O(p) \sqrt{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho(i_1 \rightarrow a_1)]} \sqrt{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho(i_1 \rightarrow \bar{a}_1)]} \quad (\text{Cauchy-Schwarz}) \\
&\leq O(p) \sqrt{\sum_{\substack{\Pi(\rho, \rho \circ \sigma'_1) = i_1 \\ \rho(i_1) = a_1}} \Pr[\rho]} \sqrt{\sum_{\substack{\Pi(\rho \circ \sigma_1, \rho) = i_1 \\ \rho(i_1) = \bar{a}_1}} \Pr[\rho]}
\end{aligned}$$

$$\begin{aligned}
&\frac{\sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \Pr[\rho]}{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho]} \\
&\quad \sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \sqrt{\Pr[\rho(i_1 \rightarrow a_1, i_2 \rightarrow a_2)] \Pr[\rho(i_1 \rightarrow a_1, i_2 \rightarrow \bar{a}_2)]} \\
&= O(p) \frac{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho(i_1 \rightarrow a_1)]}{\sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho(i_1 \rightarrow a_1)]} \\
&\leq O(p) \frac{\sqrt{\sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \Pr[\rho(i_1 \rightarrow a_1, i_2 \rightarrow a_2)]} \sqrt{\sum_{\substack{\text{NVF}(F, \rho) = i_1 \\ \text{NVF}(F, \rho(i_1 \rightarrow a_1)) = i_2 \\ F|_{\rho \circ \sigma_2} = 0, F|_{\rho \circ \sigma'_2} = 1}} \Pr[\rho(i_1 \rightarrow a_1, i_2 \rightarrow \bar{a}_2)]} \\
&\quad \sum_{\text{NVF}(F, \rho) = i_1} \Pr[\rho(i_1 \rightarrow a_1)] \\
&\leq O(p) \sqrt{\frac{\sum_{\substack{\Pi(\rho, \rho(i_1 \rightarrow a_1)) = i_1 \\ \rho(i_1) = a_1 \\ \text{NVF}(F, \rho) = i_2 \\ F|_{\rho(i_2 \rightarrow a_2) = 0, F|_{\rho(i_2 \rightarrow \bar{a}_2)} = 1}} \Pr[\rho(i_2 \rightarrow a_2)]}{\sum_{\substack{\Pi(\rho, \rho \circ \sigma'_1) = i_1 \\ \rho(i_1) = a_1}} \Pr[\rho]}} \sqrt{\frac{\sum_{\substack{\Pi(\rho, \rho(i_1 \rightarrow a_1)) = i_1 \\ \rho(i_1) = a_1 \\ \text{NVF}(F, \rho) = i_2 \\ F|_{\rho(i_2 \rightarrow a_2) = 0, F|_{\rho(i_2 \rightarrow \bar{a}_2)} = 1}} \Pr[\rho(i_2 \rightarrow \bar{a}_2)]}{\sum_{\substack{\Pi(\rho, \rho \circ \sigma'_1) = i_1 \\ \rho(i_1) = a_1}} \Pr[\rho]}}
\end{aligned}$$

Let $\mathcal{A} := \{\rho : \text{NVF}(F, \rho) = i_1\}$